

CLASS 11-CBSE MATHEMATICS

CHAPTER: SETS-SET-1

TIME : 1Hr

Mx.Marks:30

SECTION A

Q.I Answer the following questions

[1X3 = 3 MARKS]

- Write the following as intervals:
(i) $\{x: x \in \mathbb{R}, -4 < x \leq 6\}$ (ii) $\{x: x \in \mathbb{R}, -12 < x < -10\}$
(iii) $\{x: x \in \mathbb{R}, 0 \leq x < 7\}$ (iv) $\{x: x \in \mathbb{R}, 3 \leq x \leq 4\}$
- What universal set (s) would you propose for each of the following:
(i) The set of right triangles (ii) The set of isosceles triangles
- If S and T are two sets such that S has 21 elements, T has 32 elements, and $S \cap T$ has 11 elements, how many elements does $S \cup T$ have?

SECTION B

Q.II Answer the following questions

[2X5 = 10 MARKS]

- Are the following pair of sets equal? Give reasons.
(i) $A = \{2, 3\}$; $B = \{x: x \text{ is solution of } x^2 + 5x + 6 = 0\}$
(ii) $A = \{x: x \text{ is a letter in the word FOLLOW}\}$;
 $B = \{y: y \text{ is a letter in the word WOLF}\}$
- In a group of 70 people, 37 like coffee, 52 like tea, and each person likes at least one of the two drinks. How many people like both coffee and tea?
- If $A = \{x: x \text{ is a natural number}\}$, $B = \{x: x \text{ is an even natural number}\}$
 $C = \{x: x \text{ is an odd natural number}\}$ and $D = \{x: x \text{ is a prime number}\}$, find
(i) $A \cap B$ (ii) $A \cap C$ (iii) $A \cap D$ (iv) $B \cap C$
(v) $B \cap D$ (vi) $C \cap D$
- Show that for any sets A and B,
 $A = (A \cap B) \cup (A - B)$ and $A \cup (B - A) = (A \cup B)$
- In a group of students 100 students know Hindi, 50 know English and 25 know both. Each of the students knows either Hindi or English. How many students are there in the group?



SECTION C

Q.III Answer the following questions

[3X3 = 9 MARKS]

9. From the sets given below, select equal sets:

$A = \{2, 4, 8, 12\}$, $B = \{1, 2, 3, 4\}$, $C = \{4, 8, 12, 14\}$, $D = \{3, 1, 4, 2\}$

$E = \{-1, 1\}$, $F = \{0, a\}$, $G = \{1, -1\}$, $H = \{0, 1\}$

10. If $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{5, 6, 7, 8\}$ and $D = \{7, 8, 9, 10\}$; find

(i) $A \cup B$

(ii) $A \cup C$

(iii) $B \cup C$

(iv) $B \cup D$

(v) $A \cup B \cup C$ (vi) $A \cup B \cup D$

(vii) $B \cup C \cup D$

11. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$.

Find (i) A' (ii) B' (iii) $(A \cup B)'$ (iv) $(B - C)'$

SECTION D

Q.IV Answer the following questions

[4X2 = 8 MARKS]

12. In a survey of 600 students in a school, 150 students were found to be taking tea and 225 taking coffee, 100 were taking both tea and coffee. Find how many students were taking neither tea nor coffee?

13. In a survey of 60 people, it was found that 25 people read newspaper H, 26 read newspaper T, 26 read newspaper I, 9 read both H and I, 11 read both H and T, 8 read both T and I, 3 read all three newspapers. Find:

(i) the number of people who read at least one of the newspapers.

(ii) the number of people who read exactly one newspaper.

